1. Download visual studio code from
   https://code.visualstudio.com/Download
2. Refer uv-Astral Docs
   https://docs.astral.sh/uv/
3. Open powershell terminal in visual studio code and run below command to install uv
   powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
4. Make sure that uv is properly installed and installation path set to windows env variable
   $env:PATH += ";C:\Users\ADMIN\.local\bin"
5.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\manish\ejagruti\genai\myprojects\firstvsproject> uv --version
uv 0.11.3 (45da18ac3 2026-04-01 x86_64-pc-windows-msvc)
PS C:\manish\ejagruti\genai\myprojects\firstvsproject>
```

6. Now install python using uv, run below command in command line
   uv python install 3.12

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Microsoft Windows [Version 10.0.26200.8039]
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C:\manish\ejagruti\genai\myprojects\firstvsproject>uv python install 3.13
Installed Python 3.13.12 in 12.02s
 + cpython-3.13.12-windows-x86_64-none (python3.13.exe)

C:\manish\ejagruti\genai\myprojects\firstvsproject>uv python install 3.12
Installed Python 3.12.13 in 9.78s
 + cpython-3.12.13-windows-x86_64-none (python3.12.exe)
```

   uv python install 3.13

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Microsoft Windows [Version 10.0.26200.8039]
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C:\manish\ejagruti\genai\myprojects\firstvsproject>uv python install 3.13
Installed Python 3.13.12 in 12.02s
 + cpython-3.13.12-windows-x86_64-none (python3.13.exe)
```
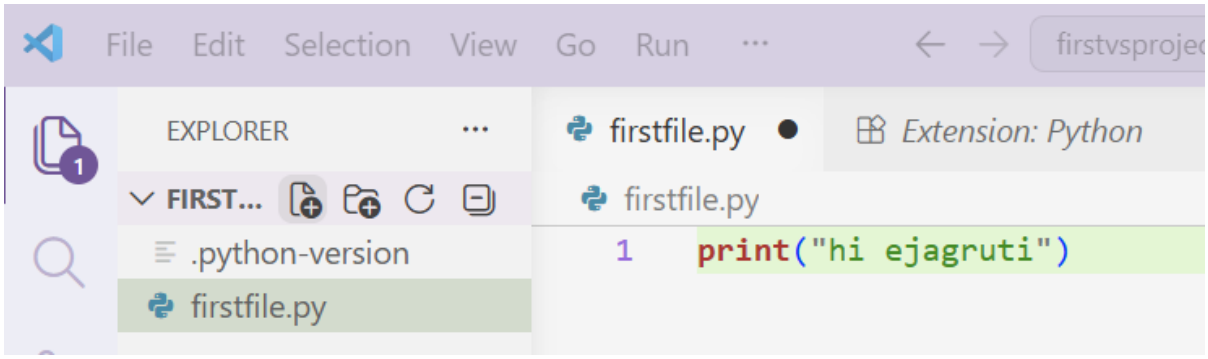

7. If you want your project to refer 3.12 then

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

C:\manish\ejagruti\genai\myprojects\firstvsproject>uv python pin 3.13
Pinned `.python-version` to `3.13`
```

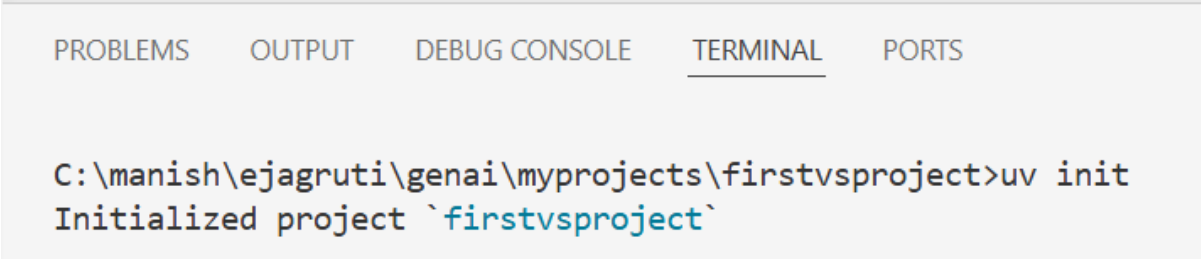
8. If you want your project to refer 3.13 then

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

C:\manish\ejagruti\genai\myprojects\firstvsproject>uv python pin 3.12
Updated `.python-version` from `3.13` -> `3.12`
```

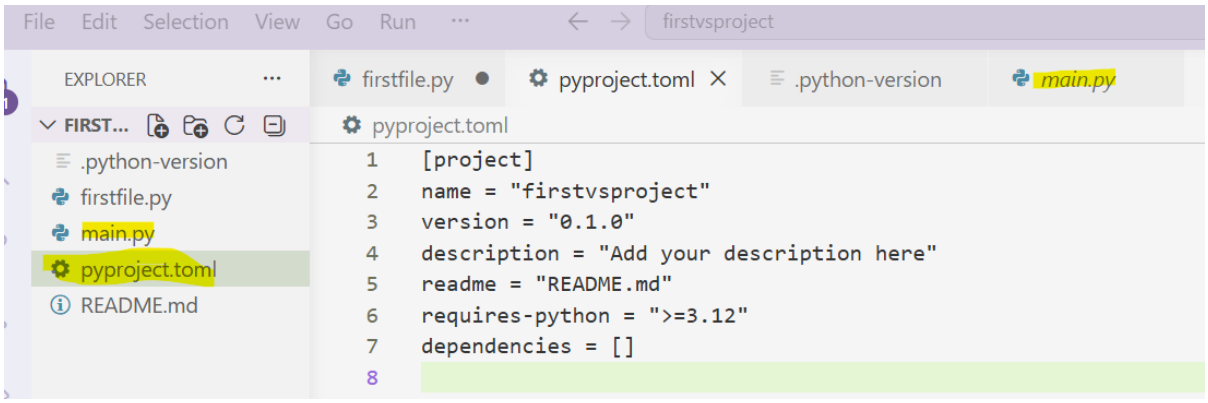
9. Now we need a virtual environment to run our python file
   Before :



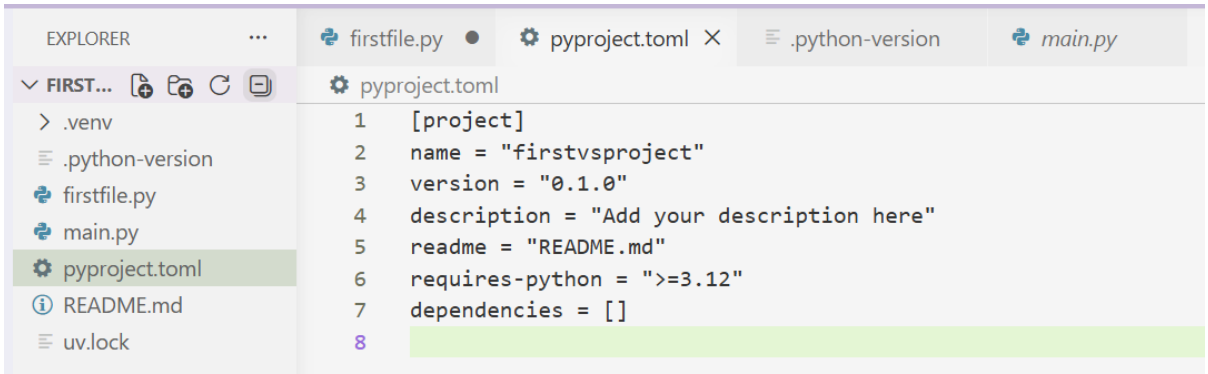
Run command : uv init



After:



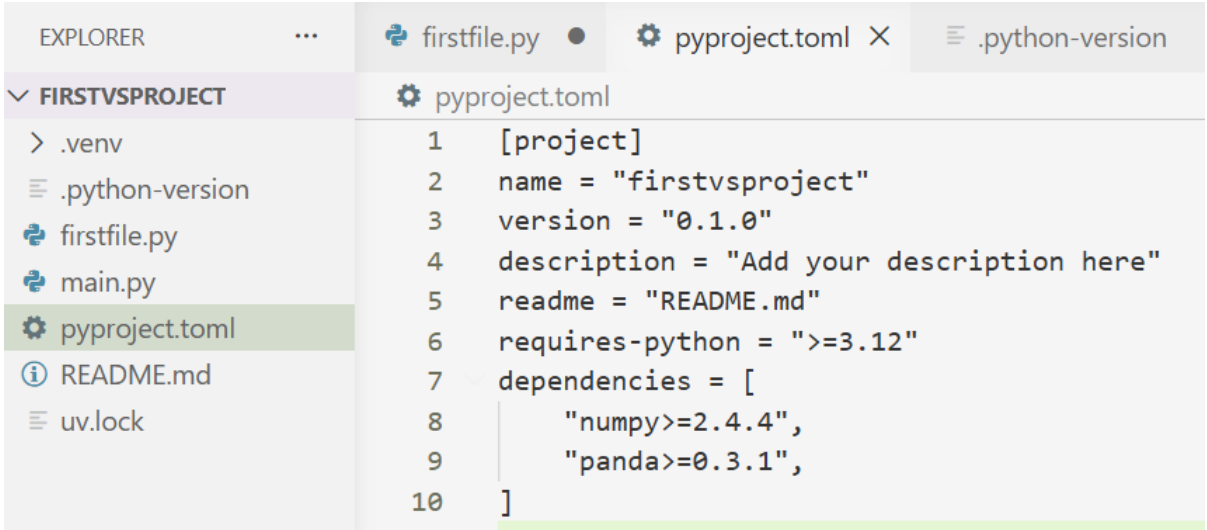
Run command : uv sync



10. To add any library run below command

```
C:\manish\ejagruti\genai\myprojects\firstvsproject>uv add numpy
Resolved 2 packages in 744ms
Prepared 1 package in 4.69s
Installed 1 package in 581ms
 + numpy==2.4.4
```

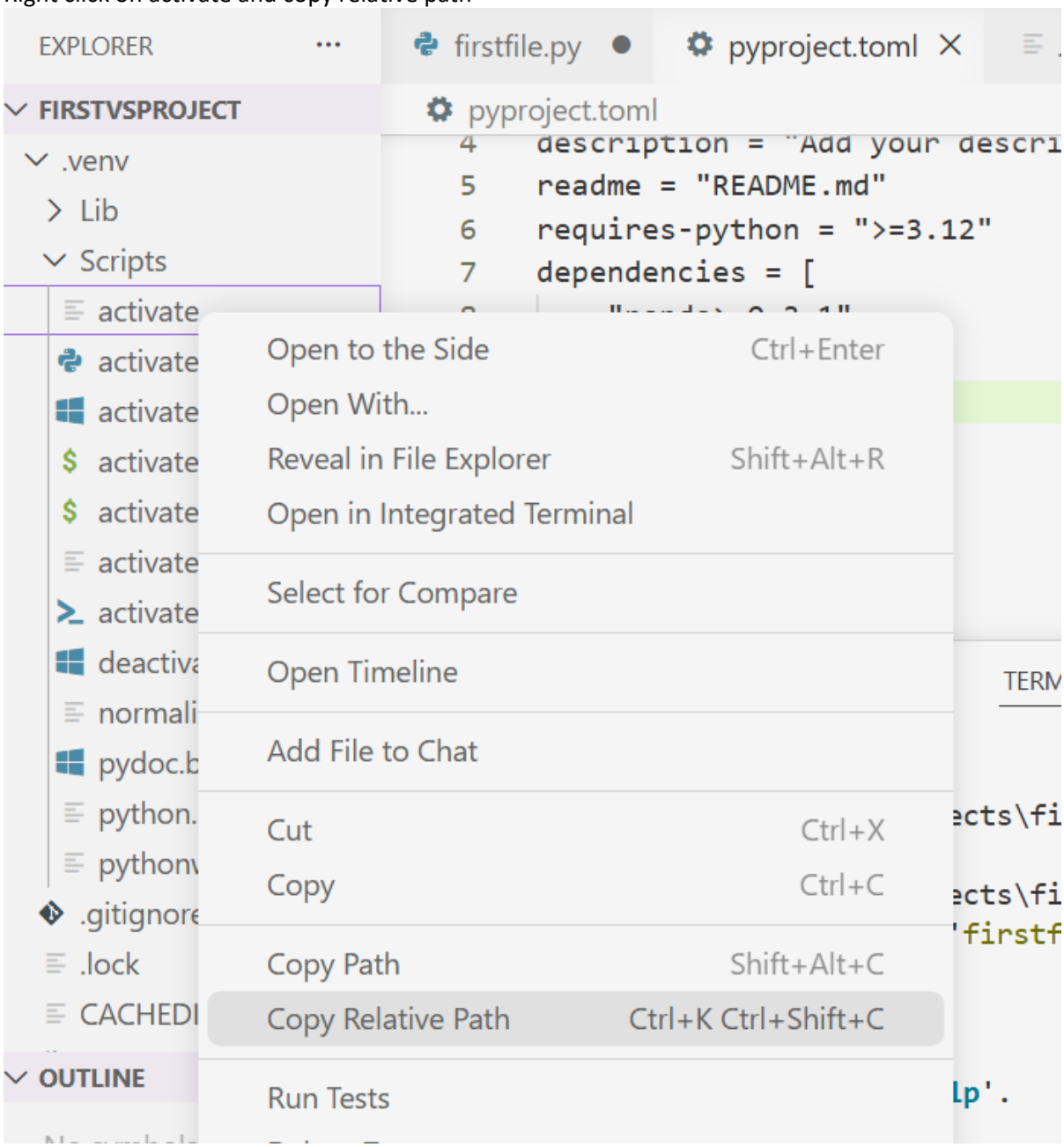
```
C:\manish\ejagruti\genai\myprojects\firstvsproject>uv add panda
Resolved 9 packages in 12.31s
      Built panda==0.3.1
Prepared 7 packages in 2.43s
Installed 7 packages in 293ms
 + certifi==2026.2.25
 + charset-normalizer==3.4.7
 + idna==3.11
 + panda==0.3.1
 + requests==2.33.1
 + setuptools==82.0.1
 + urllib3==2.6.3
```



11. To remove library : uv remove numpy

```
C:\manish\ejagruti\genai\myprojects\firstvsproject>uv remove numpy
Resolved 8 packages in 373ms
Uninstalled 1 package in 226ms
 - numpy==2.4.4
```

12. In order to Set your virtual environment :
    Right click on activate and copy relative path



13. Now paste this path in your command line

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   cmd

Microsoft Windows [Version 10.0.26200.8039]
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C:\manish\ejagruti\genai\myprojects\firstvsproject>c:\manish\ejagruti\genai\myprojects\firstvsproject\.venv\Scripts
\activate.bat

(firstvsproject) C:\manish\ejagruti\genai\myprojects\firstvsproject>.venv\Scripts\activate

(firstvsproject) C:\manish\ejagruti\genai\myprojects\firstvsproject>
```

14. Finally run below command to run your python code

```
EXPLORER        uv.lock   firstfile.py   pyproject.toml   .python-version
FIRSTVSPROJECT
  > .venv
    .python-version
    firstfile.py
    pyproject.toml
    README.md
    uv.lock

firstfile.py
1   print("hi ejagruti")

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   cmd

(firstvsproject) C:\manish\ejagruti\genai\myprojects\firstvsproject>uv run py
thon firstfile.py
hi ejagruti

(firstvsproject) C:\manish\ejagruti\genai\myprojects\firstvsproject>
```